

Project (CN6000)

Final Proposal Form

Programme: BSc (Hons) Computer Science

Year:2025

Student Number: 2576800

Proposed Title: Personalized nutrition recommendations using explainable AI and Gut - microbiome Data

Proposed Aim: Developing a machine learning model that connects health information and gut microbiota to provide specific nutritional guidance.

Proposed Objectives:

By the end of this project, I will be able:

1. To analyse and thoroughly assess cutting-edge Explainable AI (XAI) models (such as LIME, SHAP, and attention mechanisms) that are appropriate for understanding complex biological data, with a focus on their use in the microbiome of the gut analysis and personalised diet.
2. To Research and define data requirements and processing methods to integrate gut microbiome data with personal health metrics.
3. To conduct an initial analysis utilising an existing public or curated dataset that includes associated gut microbiome and dietary or health-related data, which will function as the training and validation set for the proposed AI model.
4. To implement quantitative methods to organised surveys, rating scales, and statistical analysis to find out how clear, satisfied, and reliable AI-generated nutrition advice is and how well it works with user feedback.
5. To implement a user-friendly interface that offers personalised nutritional advice along with explicit explaining reasoning generated by the XAI model.
6. Evaluate system results through assessing AI recommendations' ethical transparency, usability, and accessibility.
7. Analyse system results to improve performance and guide future developments in nutrition-related explainable AI.

Rationale:

The evolution towards personalised nutrition, especially using complicated gut microbiome data, promises to make health much better. But the fact that many powerful AI models are "black boxes" by nature can make it hard for them to be used in sensitive areas like health. Generic nutrition advice doesn't work, and even though AI can make it more personal, general suggestions can make people not trust them, not follow them, and not understand why a certain diet is suggested.

Explainable AI (XAI) is used in this project to address this crucial gap. An XAI system can offer clear clarifications and insights into how the gut microbiome profile and other health data result in particular suggestions for diets.

- Increase User Trust and Safety: Users are more likely to follow advice that makes sense to them and connects to their own biological data.

- Assist People Make Smart Choices: People can learn more about their body's unique needs and make smart choices about what they eat.
- Support HealthCare Specialists: Healthcare providers can learn more about why - recommendations are made, which helps them teach and care for patients.
- Ensure that the model is accurate to identify any errors: Explainable AI techniques can help look closely at how the model makes decisions, making sure they are fair and finding any biases that might not be intentional.

An important step towards genuinely assisting people with accurate, transparent, and reliable nutritional advice is the creation of an Explainable AI system for personalised nutrition using gut microbiome data, improving personalised healthcare and avoiding illness.

Facilities required:

- High-performance resource utilisation (for data processing, XAI calculations, and training AI models). Access to pertinent datasets (such as dietary survey data, physiological markers, public gut microbiome datasets, or simulated data in the event that real data is impractical).
- specialised software (such as QIIME2, R, and Python with scikit-learn, TensorFlow/PyTorch, SHAP, and LIME) for bioinformatics analysis and advanced machine learning.
- Additionally, to develop an engaging system that can present clear explanations and personalised recommendations in an easily understandable format, user interface development tools (like a web or mobile app framework) will be required. Computational scalability can be achieved by using cloud computing platforms such as AWS or Google Colab, among others.
- These combined resources will ensure the successful development, testing, and deployment of the Explainable AI-based personalized nutrition recommendation system.

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